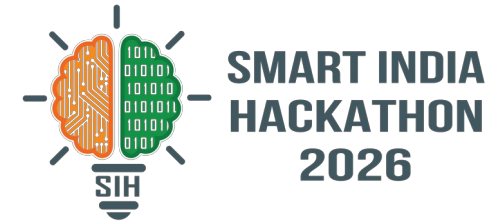


SMART INDIA HACKATHON 2026



MausamSetu

Problem Statement ID: SIH26081

Problem Statement Title

Hybrid AI–NWP Multi-Model
Forecast Blending System

Theme: Disaster Management

PS Category: Software

Team ID: 189879

Team Name: HEXATECH



Proposed solution

Combine physical, ensemble and AI forecasts, then verify which blend works for each context.

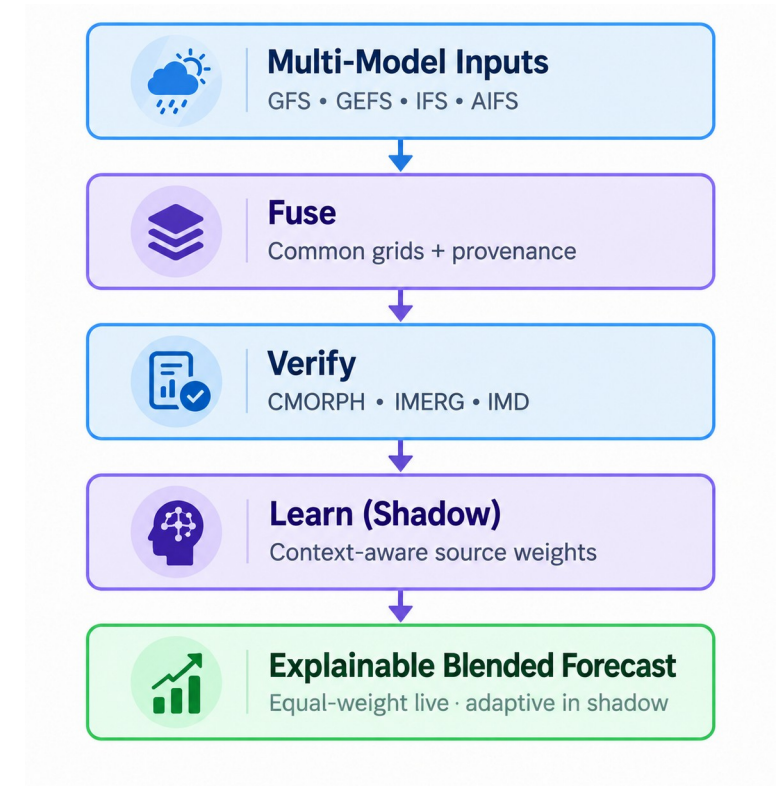
How it addresses the problem

GFS, GEFS, IFS and AIFS share consistent grids and valid times. CMORPH, IMERG and IMD provide separate verification scorecards.

Innovation and uniqueness

Learn context-dependent weights in a shadow workflow. Handle unavailable sources explicitly. Preserve each run, lead, unit and checksum for inspection.

Live: equal-weight baseline. Adaptive candidates await acceptance.



Methodology and implementation

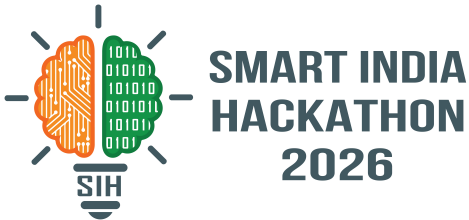
Stage	Implementation
Ingest and align	Validate run, lead, units and grids for GFS / GEFS / IFS / AIFS.
Verify independently	Compare exact valid windows with CMORPH, IMERG and IMD.
Learn in shadow	Train by chronology; test by event block, lead and variable.
Handle missing sources	Set unavailable-source weights to zero; renormalize valid weights.
Publish and inspect	Serve live baseline maps, source comparisons and provenance.

Technology stack

Python / FastAPI • PostgreSQL / PostGIS • xarray / ecCodes
Next.js / React / MapLibre • NumPy / SciPy • Docker Compose



FEASIBILITY AND VIABILITY



Feasibility demonstrated

Deployed: Vercel frontend and OCI API/workers. Four live forecast sources, verification pipelines and MOSDAC rainfall context. Tailscale protects SSH.
9 IMD reports through 30 Sep 2026, 03:00 UTC. Adaptive acceptance remains pending.

Potential challenges and mitigation

Challenge / risk	Strategy
Late or missing model data	Track freshness and availability; expose degraded-source status.
Different grids and time windows	Normalize units and coordinates; verify matching accumulation windows.
Limited extreme-event evidence	Collect independent prospective cases; retain the baseline until gates pass.
NCUM / latest-cycle NEPS access	Use open sources now; integrate authorized Indian feeds when available.

Next: off-site restore drill, external alerts, institutional identity and scientific review.

Potential impact on the target audience

Forecasters

Inspect competing forecasts and evidence by location, lead time and variable.

Disaster managers

Use traceable forecast context alongside official warnings for preparedness.

Farmers and institutions

Support weather-sensitive planning through reviewed, locally validated guidance.

Benefits of the solution

Social

Clear source provenance and accessible preparedness resources.

Economic

Open-data prototype on modest hardware; less manual comparison of model outputs.

Environmental

Potential support for flood, heat and wind preparedness after local validation.

Expected benefits; no measured reduction in losses or validated operational advantage claimed.

Historical temperature result: 10.9% lower CRPS

WeatherBench 2 IFS-HRES + Pangu against ERA5, India-region bounding box.
Adaptive 0.5494 °C vs equal mixture 0.6164 °C. 36 held-out 2022 initializations.

Retrospective temperature experiment. ERA5 is reanalysis. Results do not establish operational rainfall, wind or extreme-event superiority.

Preliminary rainfall shadow evaluation: +24 h

GFS / GEFS / IFS / AIFS. CRPS against CMORPH: 1.737 vs 1.811 (4.1% lower).
IMERG cross-reference: 1.549 vs 1.630 (5.0% lower). Adaptive vs equal weighting.

Only 2 held-out event blocks. Experimental candidate, not deployed.

Lower CRPS indicates better probabilistic performance. Production gates remain unmet.

References: NOAA / ECMWF / WeatherBench 2 / CMORPH / NASA IMERG / IMD / MOSDAC

Full source links and methodology: <https://mausamsetu-mauve.vercel.app/sih>

Live prototype and evidence: <https://mausamsetu-mauve.vercel.app/sih>

Source code: <https://github.com/Jaydeng75/mausamsetu>